

Max Tacker

tacker.maxwell@gmail.com | Active Security Clearance

Skills and Knowledge

Languages: C++, C, C#, Bash, Python, Ada, MATLAB

Development Tools: Git, GDB, AdaMulti, Simulink Embedded Coder, Rational DOORS

OS & RTOS: Linux, Green Hills Integrity-178 RTOS

Buses & Interfaces: MIL-STD-1553/1760, ARINC 429, CAN, Ethernet, RS-232, SPI, I2C, UART

Networking: TCP/IP, UDP, Link 16

Aircraft Certification: DO-178C, DO-254, MIL-STD-810H

Experience

Senior Software Engineer III, Lockheed Martin Corporation – Fort Worth, TX August 2024 – Present

- Delivered application releases ahead of schedule and under budget as the lead engineer of 10-developer team tasked with integrating advanced stores on the F-35 platform.
- Directed an engineering team optimizing data-link sensor fusion algorithms for throughput-constrained legacy hardware.
- Spearheaded successful multi-system integration events with international partners under compressed timelines.
- Managed concurrent roles as Scrum Master, Product Owner, and Technical Lead across multiple software development streams.
- Authored safety-critical, object-oriented C++ software deployed on Green Hills Integrity-178 RTOS and Embedded Linux.
- Integrated novel weapons systems and sensors utilizing MIL-STD-1553 and Ethernet communications.
- Integrated AI-driven tooling into development workflows to accelerate interface design, technical documentation retrieval, and test generation.
- Regularly interfaced with program leadership, external stakeholders, and vendors to represent the Fire Control and Stores Management software group.

Senior Flight Controls Software Engineer II, Bell Textron, Inc – Fort Worth, TX January 2022 – August 2024

- Developed, integrated, and validated embedded flight control software for multiple clean-sheet fly-by-wire aircraft platforms using C, MATLAB, and Simulink Embedded Coder.
- Developed embedded software for crew stations, flight control laws, actuation devices, avionics, air data, and engine controllers.
- Directed flight test scenarios for pilots in a hardware-in-the-loop systems integration simulator to validate flight control software.
- Managed full lifecycle requirements development and verification in compliance with DO-178C standards.
- Refactored system requirements to optimize safety-critical Flight Control Computer (FCC) interfaces.
- Executed system safety assessments and failure hazard analyses to enforce airworthiness standards.

Systems Engineer I – II, Raytheon – Dallas, TX June 2019 – January 2022

- Architected multi-threaded automated test software in C, C++, and NI TestStand for mission computer hardware evaluation.
- Designed, validated, and maintained test suites for acceptance and environmental stress testing of units under test (UUTs).

Education and Certifications

Baylor University, Waco, TX – BS in Electrical and Computer Engineering (Minor: Mathematics) May 2019

Certifications: Private Pilot Certificate | NCEES FE Electrical and Computer